

WASP-36b

R-1972

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_150307 · created 2026-08-07T00:02:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457088.755 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.73 +/- 1.6 [%]
Orbital Inclination (inc)	87.54 +/- 2.83
Airmass coefficient 1 (a1)	0.58 +/- 0.36
Airmass coefficient 2 (a2)	0.28 +/- 0.98
Scatter in the residuals of the lightcurve fit is	57.36 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	5.25
Transit Duration (day)	0.0924 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.048 vs 0.1368 (deviation 0.56σ)
Duration fit vs published	0.0924 d vs 0.0758 d (deviation 1.71σ)
Mid-time O–C	-0.2 min
Depth SNR	3.3
Residual scatter	57.36 %

Light curve

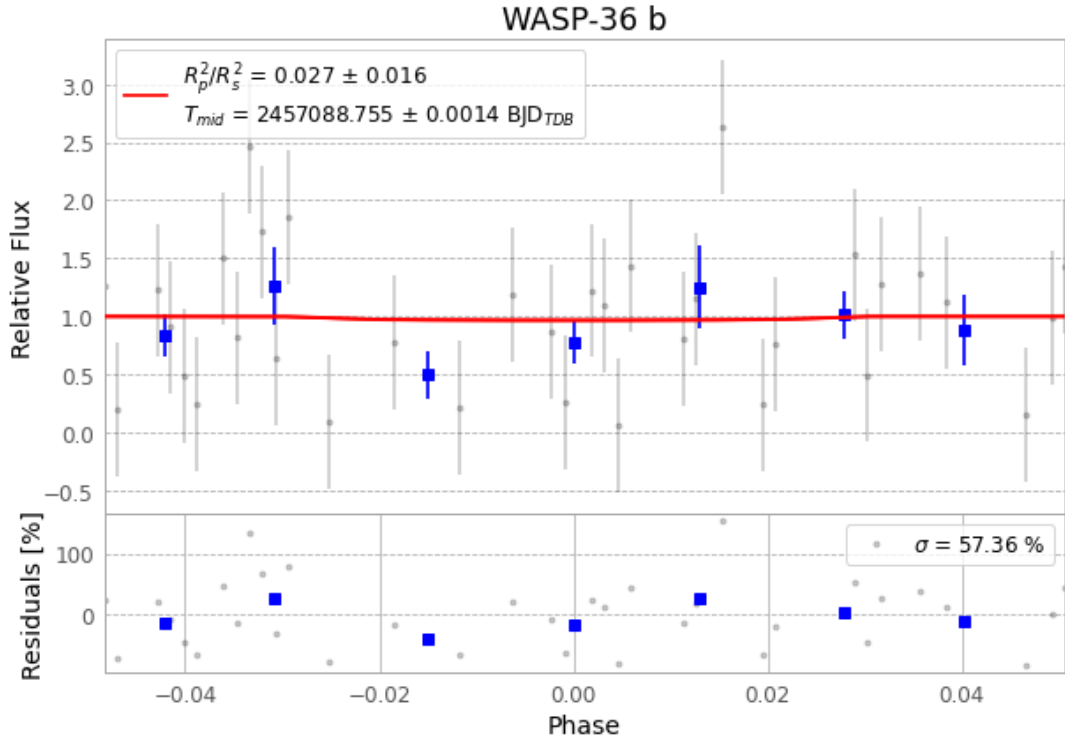


Figure 1 — FinalLightCurve_WASP-36 b_2015-03-06.png (SHA-256 a03bc8e27bb04149...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 378], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d7219a12d4ac8f4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

76 FITS frames (50 MB), WASP-36150307041212.FITS → WASP-36150307080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150307.log (ce691c8768dc...), FinalLightCurve_WASP-36 b_2015-03-06.png (a03bc8e27bb0...), FinalLightCurve_WASP-36 b_2015-03-06.csv (90a4b6474f13...)

Compute resources

Wall time 132.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:02Z.